

SmolVLM

Redefining Small and Efficient Multimodal Models

Hugging Face · Stanford University

Competitive multimodal performance at sub-1GB RAM | 256M – 2.2B parameters | Fully open-source

256M

Smallest Model

<1 GB

GPU RAM

300×

Smaller than Llama-3.1-8B

13

Benchmarks Evaluated

Large VLMs Are Powerful but Impractical for Edge Deployment

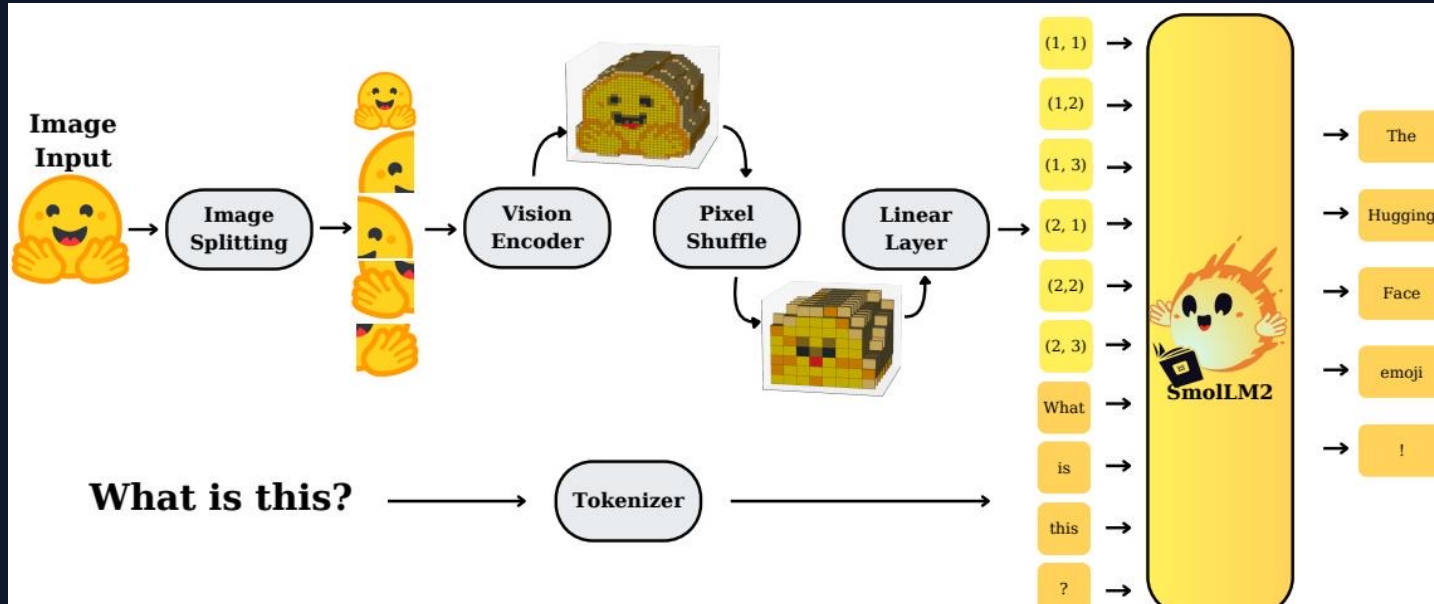
- ▶ State-of-the-art VLMs (7B–80B params) require 14–80+ GB GPU RAM
- ▶ Mobile/edge devices have 4–16 GB total RAM — no room for large VLMs
- ▶ Cloud inference adds latency, cost, and privacy concerns
- ▶ Existing efficient models still consume 27+ GB (e.g., Molmo-A1B-7B: 27.7 GB)
- ▶ No sub-1B parameter VLM with competitive multimodal performance existed

RAM COMPARISON

Idefics-80B	80B params	~160 GB
Molmo-A1B-7B	7B params	27.7 GB
InternVL2-2B	2B params	~8 GB
SmolVLM-2.2B	2.2B params	4.9 GB
SmolVLM-500M	500M params	1.2 GB
SmolVLM-256M	256M params	0.8 GB
SmolVLM-256M: sub-1GB inference on smartphones		

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2

Backbone



1 IMAGE SPLITTING

Input images split into sub-images for granular encoding

2 VISION ENCODER

SigLIP encodes sub-images into visual feature maps

3 PIXEL SHUFFLE

Space-to-depth rearrangement compresses tokens by r^2

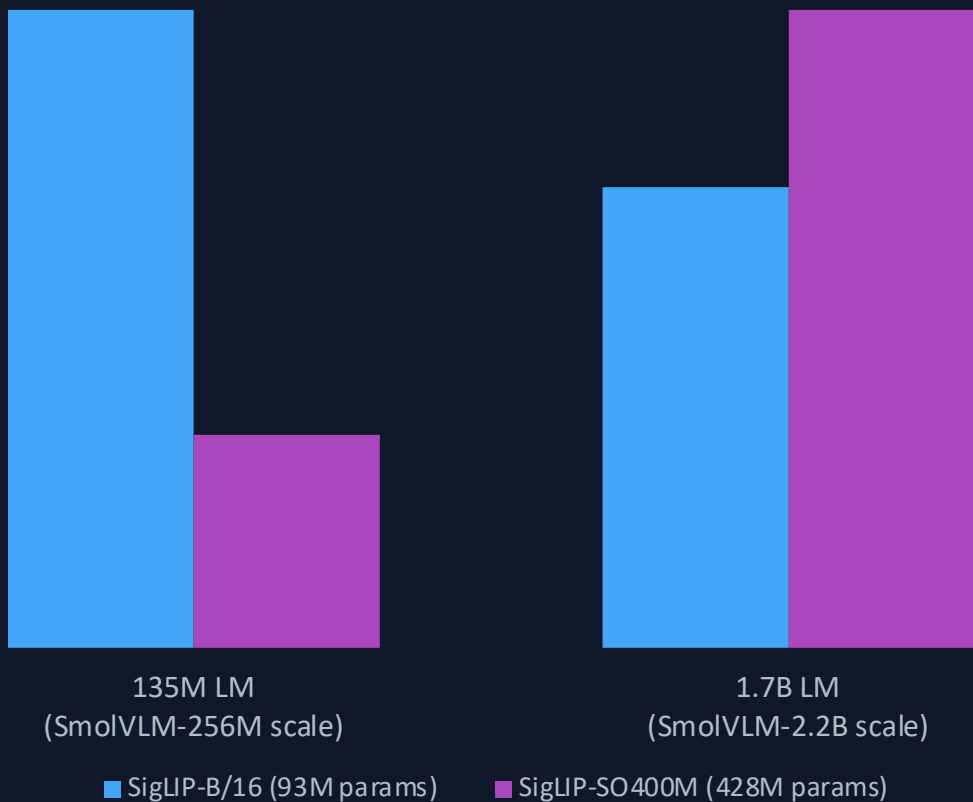
4 LINEAR PROJECTION

Compressed features projected into LM embedding space

5 SmolLM2 BACKBONE

Language model processes concatenated visual + text tokens

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs



93M SigLIP-B/16

Outperforms the 428M SigLIP-SO400M encoder when paired with 135M language model

4.6× Fewer Params

93M vs 428M vision encoder — massive compute savings with better downstream accuracy

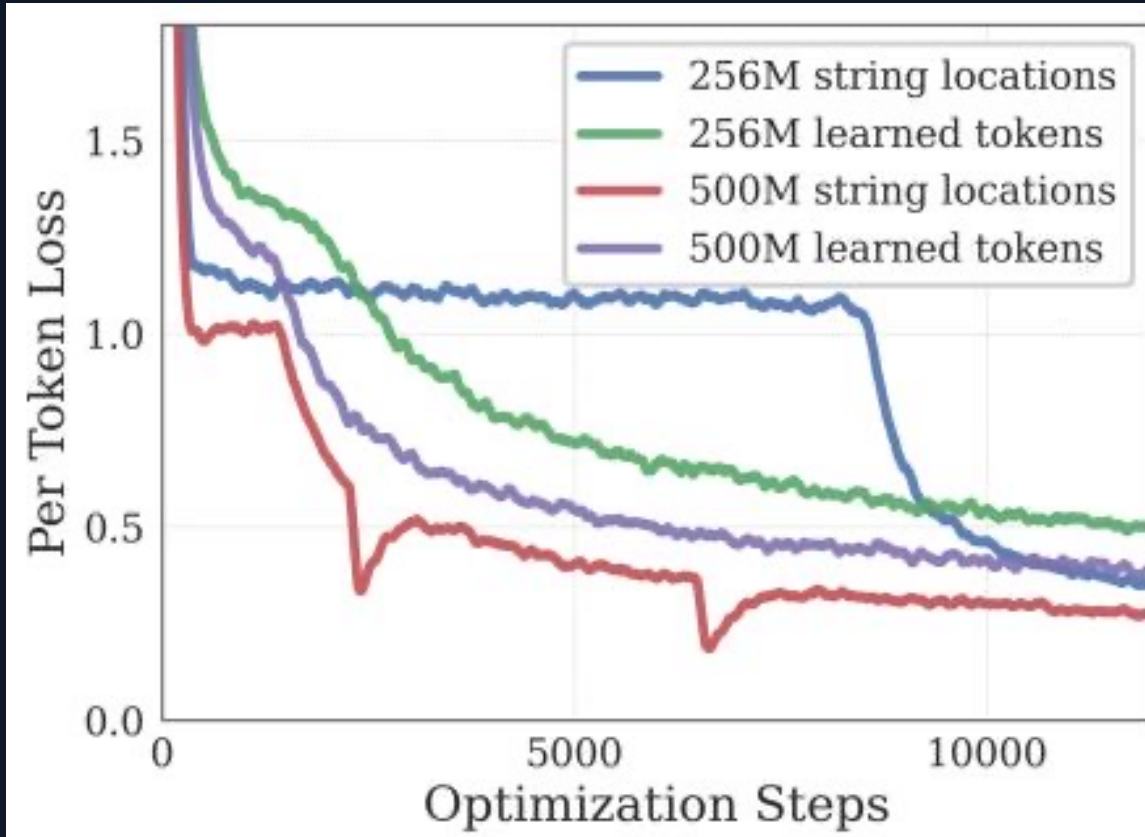
Crossover at 1.7B

Larger encoder only justifies its 10% parameter overhead at 1.7B+ LM scale

Design Principle

Match encoder capacity to LM capacity — oversized encoders waste compute in small VLMs

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models



PIXEL SHUFFLE: SPACE-TO-DEPTH TOKEN COMPRESSION

r = 1

No reduction

729 tokens per sub-image

r = 2

4× reduction

~182 tokens per sub-image

r = 4

16× reduction

~46 tokens per sub-image

KEY INSIGHT

Small VLMs benefit from r=4 (16× compression) — eases attention overhead and improves long-context modeling. Larger models (2.2B+) prefer r=2 for finer spatial resolution.

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

2K TOKENS

Baseline context length

Limited image resolution and detail

8K TOKENS

SmolVLM-256M / 500M

4× more context for visual tokens

16K TOKENS

SmolVLM-2.2B

8× baseline — high-res multi-image support

- ▶ RoPE base frequency increased from 10,000 to 273,000 to support longer sequences
- ▶ Fine-tuned on mixed long/short-context data to maintain short-sequence performance
- ▶ Performance improves significantly when extending context from 2K to 16K tokens
- ▶ Enables processing of higher-resolution images and multi-image inputs
- ▶ Smaller variants (256M, 500M) use 8K context — optimal balance of capacity and overhead

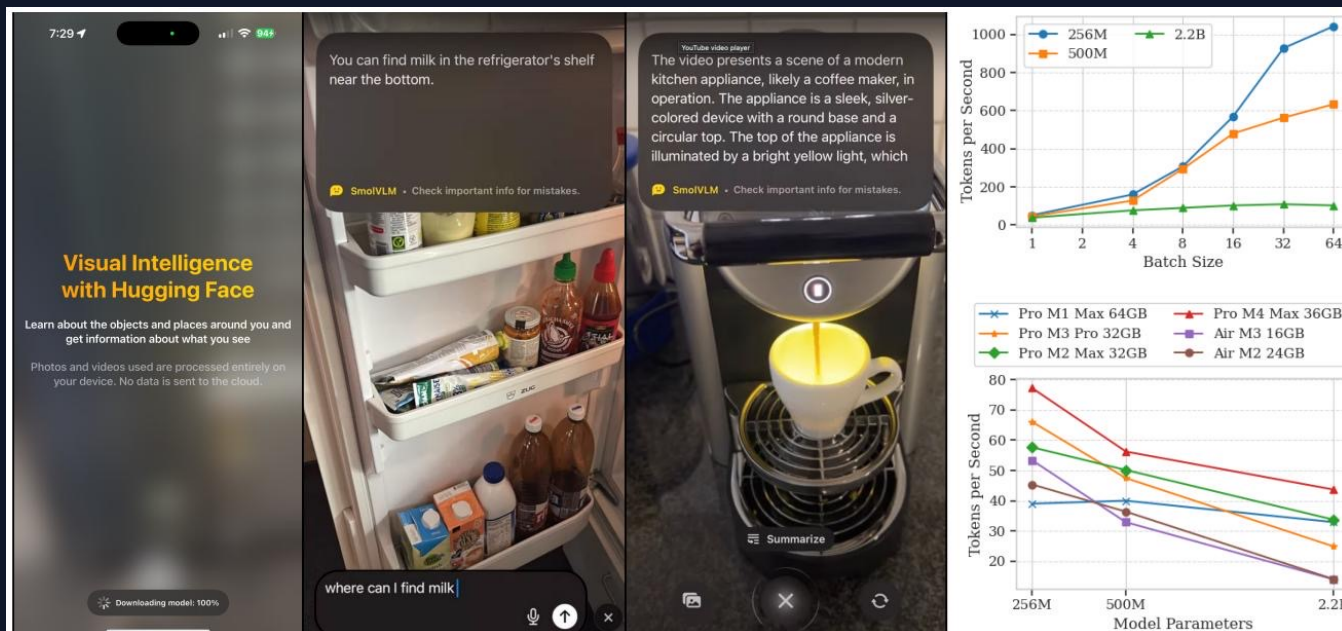
SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB

RAM

Category	Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo-A1B-7B)
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo-A1B-7B)
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo-A1B-7B)
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo-A1B-7B)
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo-A1B-7B)
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo-A1B-7B)
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo-A1B-7B)
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo-A1B-7B)
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo-A1B-7B)
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2-2B)
Video	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2-2B)
Average	All Benchmarks	44.0%	51.0%	59.8%	—

RAM Usage: 256M → 0.8 GB | 500M → 1.2 GB | 2.2B → 4.9 GB | Molmo-A1B-7B → 27.7 GB SmolVLM-256M surpasses the 300× larger Idefics-80B

Real-Time Inference on Apple Silicon: Up to 80 Tokens/Second



ON-DEVICE PERFORMANCE

~80 tok/s

SmolVLM-256M on
MacBook Pro M1 Max 64GB

~55 tok/s

SmolVLM-256M on
MacBook Air M3 16GB

~1000 tok/s

SmolVLM-256M at batch=64
on high-end Apple Silicon

- ▶ HuggingSnap iOS app — fully local inference, no cloud dependency
- ▶ Throughput scales roughly linearly with batch size for 256M model
- ▶ Enables real-time photo and video understanding on consumer hardware

Design Principles for Efficient Multimodal Models

1

BALANCE ENCODER-LM COMPUTE

Match vision encoder capacity to LM scale. 93M encoder beats 428M for sub-500M LMs.

2

EXTEND CONTEXT LENGTH

Scale from 2K to 8K–16K tokens via RoPE base tuning. Unlocks high-res and multi-image inputs.

3

AGGRESSIVE TOKEN COMPRESSION

Pixel shuffle at $r=4$ gives 16× reduction for small models. Eases attention overhead significantly.

4

SUB-1GB INFERENCE IS ACHIEVABLE

SmolVLM-256M runs at 0.8 GB RAM. Outperforms the 300× larger Llafix-80B.

5

FULLY OPEN-SOURCE RELEASE

All weights, datasets, code, and the HuggingSnap mobile app are publicly available.